

AI 모델 성능 개선 결과

예측 구간 별 특징

예측 잘 맞는 구간

- 주로 20.08.02
- 단봉형 (피크 1개)
- 상승/하강이 비교적 매끈한 이벤트
- 6시간 윈도우 내에 주요 상승이 다 들어오는 이벤트
- 기저수위 (예측 시작 직전 수위) 가 안정적인 이벤트
- 선행강우 - 수위 상태가 “ 학습에서 흔히 본 ” 범위
- 이때 Lag (지연시간) $\approx 0 \sim \pm 20$ 분 ,
Amplitude (피크) 도 대체로 0.95 ~ 1.05 근처로 유지

예측이 잘 틀리는 구간

- 다봉형 (피크 2개 이상)
- 플레토 (계단식)
- 급격한 상승
- 기저수위 (예측 시작 직전 수위) 가 이벤트 중간에 비정상적으로 이동
(배수 / 펌프 / 지류 · 역류 영향)
- 상태가 다른데 같은 강우 - 수위 반응을 강요 하는 상황 : 동일 강우라도 수위 반응이 달라짐
(선행 포화 / 저류 , 관거 / 우수토실 , 하천 시설물 운영 , 역수위 영향 등)

학습 조건 차이

Parallel LSTM

- **smoothing** : 궁내교 (1) 대곡교 (2)
- **그래프 스케일 고정** : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)
- **learning_rate** = 0.0005
- **Loss_function** = MAE
- **Seq_length** = 36 (6시간)
- **Num_of_layer** = 18
- **Dense** = 36 (1번 처리)

단일 LSTM

- **그래프 스케일 고정** : 궁내교 (0.7 - 3) 대곡교 (1.5 - 5.5)
- **learning_rate** = 0.0003
- **Loss_function** = MSE
- **dropout** = 0.0
- **Seq_length** = 36 (6시간)
- **Dense** = 1 (36번 처리)
- **학습 이벤트 순서** shuffle

관심수위 학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

- 기본

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

- 누적 강우 추가

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교 _Ti	궁내교 _Ti	대곡교 누적강우	궁내교 누적강우
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0	0.0	0.0

학습 : 총 67개 中 61개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 6개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2014 ~ 20230629 (43개)
- 검증 : 20230713 ~ 250813 (12개)
- 테스트 : 2550916 ~ 251006 (6개)

2020년 이후 관심수위 학습데이터

홍수통제소 API 호출 → MIET 활용 강우 이벤트 추출 → 관심 수위 이상 추출

- 기본

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교_Ti	궁내교_Ti
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0

- 누적 강우 추가

time	서울시 (대곡교)	성남시 (성남북초교)	광주시 (남한산초교)	성남시 (대장동)	성남시 (구미초교)	성남시 (한국학중앙연구원)	성남시 (궁내교)_WL	성남시 (궁내교)_Q	서울시 (대곡교)_WL	서울시 (대곡교)_Q	대곡교 _Ti	궁내교 _Ti	대곡교 누적강우	궁내교 누적강우
2014-07-24 15:30:00	0.0	0.0	0.0	0.0	0.0	0.0	1.12	5.11	2.17	40.56	0.0	0.0	0.0	0.0

학습 : 총 40개 中 35개 데이터 파일 사용

테스트 : 학습 데이터에서 제외한 5개 데이터 파일 (4개 이벤트)

+ 학습에서 검증에 사용된 2개 이벤트 사용

- 학습 : 2020 ~ 20240920 (25개)
- 검증 : 20240921 ~ 250919 85번 (7개)
- 테스트 : 250919 86번 ~ 251006 (3개)

관심수위
궁내교

관심수위 궁내교

• Parallel LSTM

- 1시간 : -0.54 (72.0 %)
1.94 - 1.4
- 1시간 : -0.41 (75.7 %)
1.67 - 1.08

관심 수위 이상				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9656	0.0072	0.0558	0.0852
22.07.12	0.4201	0.0230	0.1139	0.1516
23.07.11	0.8622	0.0385	0.1486	0.1963
24.07.23	0.7308	0.0217	0.0962	0.1475
25.07.16	0.7947	0.0205	0.1047	0.1435
25.08.14	0.5087	0.0346	0.1446	0.1861

관심 수위 이상 + 누적강우

* 몇시간 뒤 시점 : 오차 (유사율)

관측수위 - 예측수위

관심 수위 이상 + 누적강우				
누적	궁내교			
Test data	r2	mse	mae	rmse
20.08.02	0.8893	0.0233	0.1222	0.1529
22.07.12	-0.6973	0.0673	0.2298	0.2595
23.07.11	0.9031	0.0270	0.1052	0.1646
24.07.23	0.8385	0.0130	0.0713	0.1142
25.07.16	0.7659	0.0234	0.1271	0.1532
25.08.14	0.6406	0.0253	0.1262	0.1591

- 3시간 : -0.38(82.6 %)
2.16 - 1.78
- 1시간 : -0.52(73.1 %)
1.94 - 1.42
- 1시간 : -0.25 (84.9 %)
1.67 - 1.42

• 단일 LSTM

- 1시간 : -0.36(81.4 %)
1.94 - 1.58

관심 수위 이상				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9508	0.0104	0.0757	0.1020
22.07.12	0.0509	0.0377	0.1791	0.1941
23.07.11	0.9096	0.0253	0.1299	0.1590
24.07.23	0.7689	0.0187	0.0785	0.1367
25.07.16	0.5051	0.0496	0.1722	0.2228
25.08.14	0.7659	0.0165	0.0899	0.1285

관심 수위 이상 + 누적강우				
누적	궁내교			
Test data	r2	mse	mae	rmse
20.08.02	0.9176	0.0174	0.0993	0.1320
22.07.12	0.4645	0.0212	0.1331	0.1458
23.07.11	0.9119	0.0247	0.1283	0.1570
24.07.23	0.7989	0.0163	0.0600	0.1276
25.07.16	0.4844	0.0517	0.1744	0.2274
25.08.14	0.7933	0.0146	0.0833	0.1207

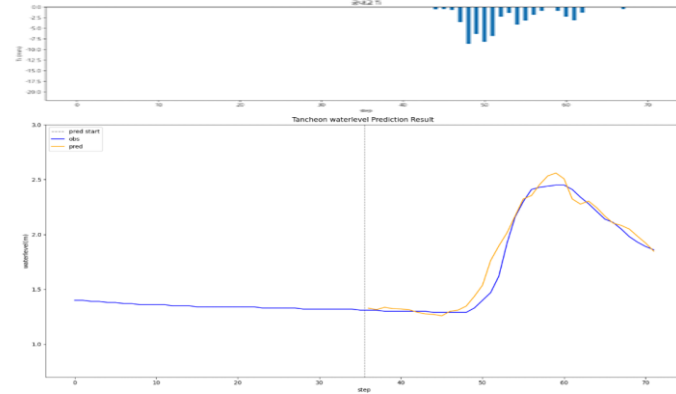
- 1시간 : -0.36(81.3 %)
1.94 - 1.58
- 1시간 : -0.22(82.1 %)
1.24 - 1.02
- 3시간 : -0.31(82.8 %)
1.79 - 1.48

궁내교 – Parallel LSTM

관심 수위 이상

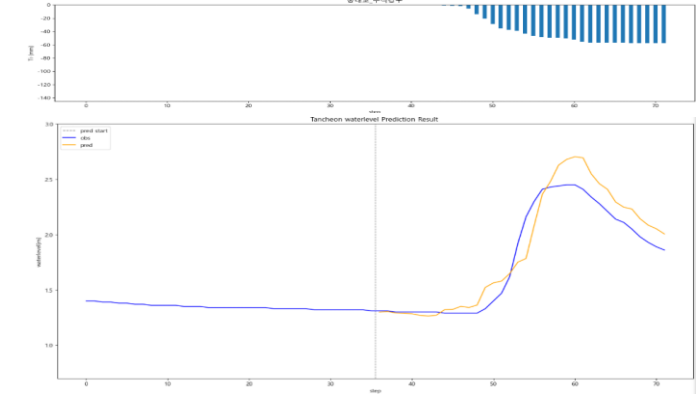
testdata_200802

R2 : 0.9656



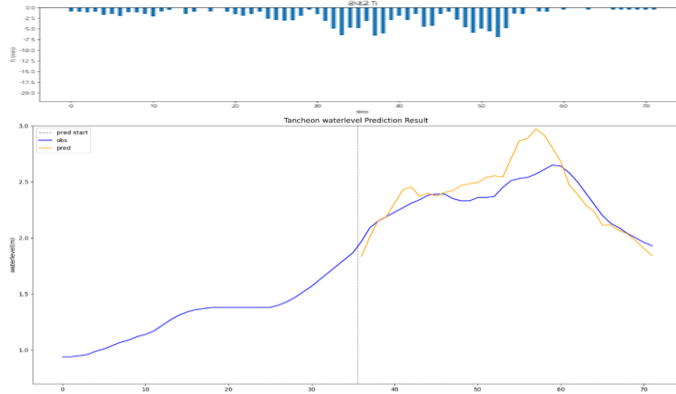
testdata_200802

R2 : 0.8893



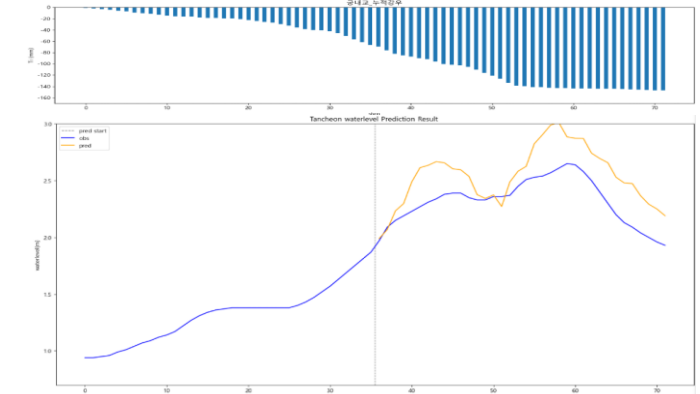
testdata_220712

R2 : 0.4201



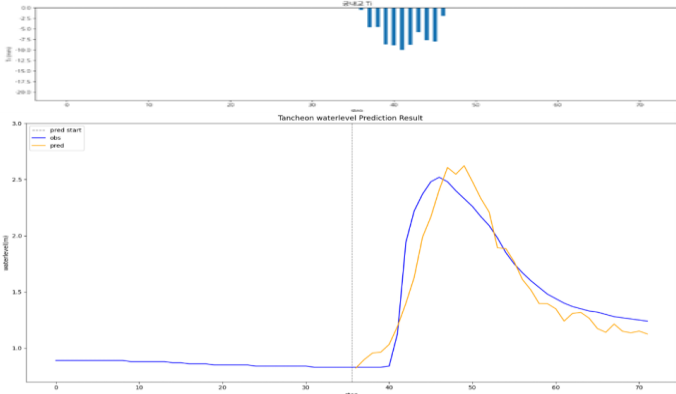
testdata_220712

R2 : -0.6973



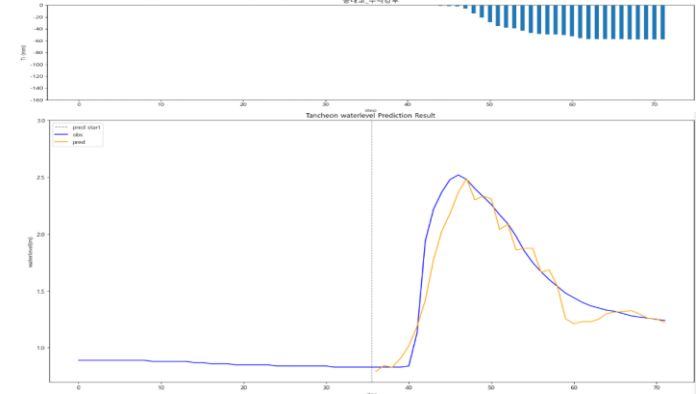
testdata_230711

R2 : 0.8622



testdata_230711

R2 : 0.9031

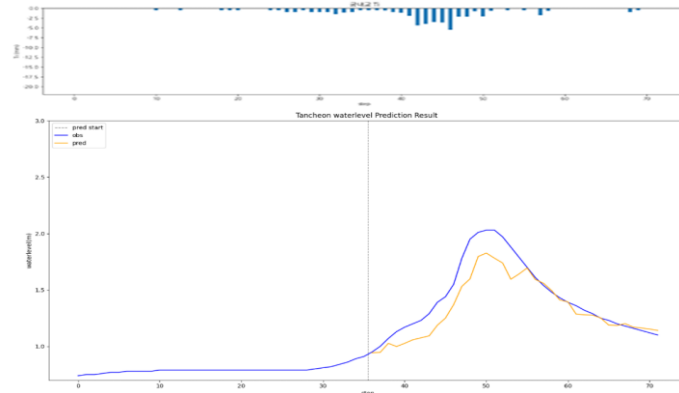
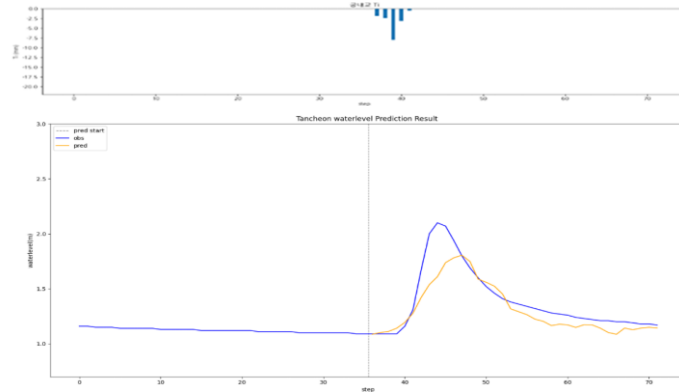


궁내교 – Parallel LSTM

관심 수위 이상

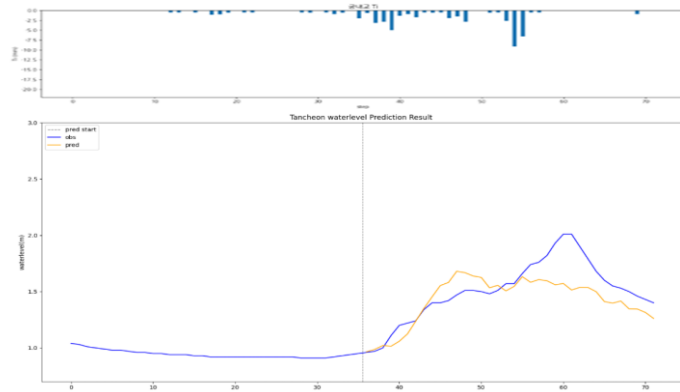
testdata_240723

R2 : 0.7308



testdata_250716

R2 : 0.7947



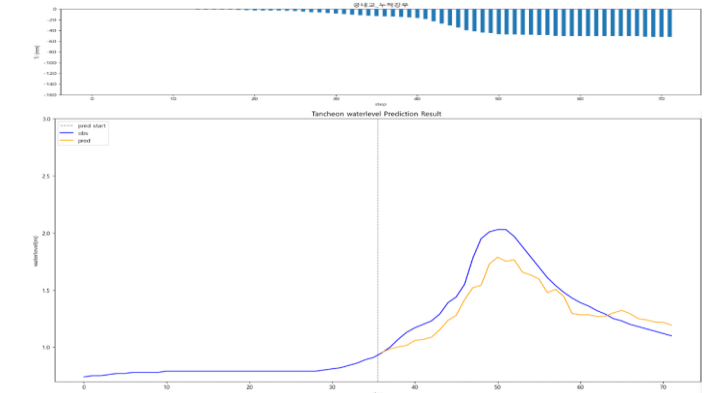
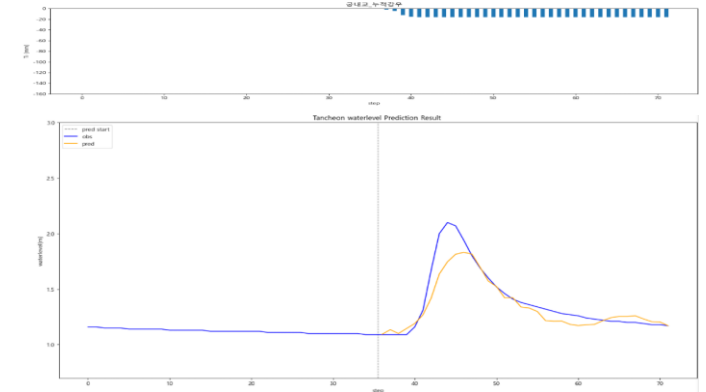
testdata_250814

R2 : 0.5087

관심 수위 이상 + 누적강우

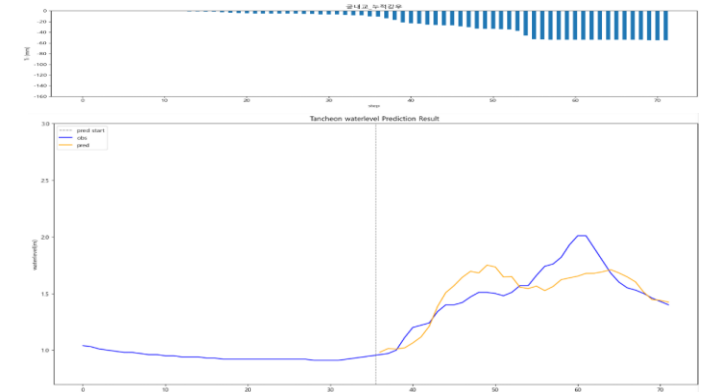
testdata_240723

R2 : 0.8385



testdata_250716

R2 : 0.7659



testdata_250814

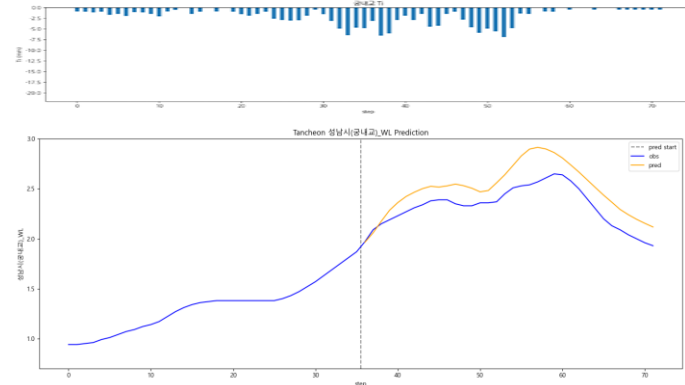
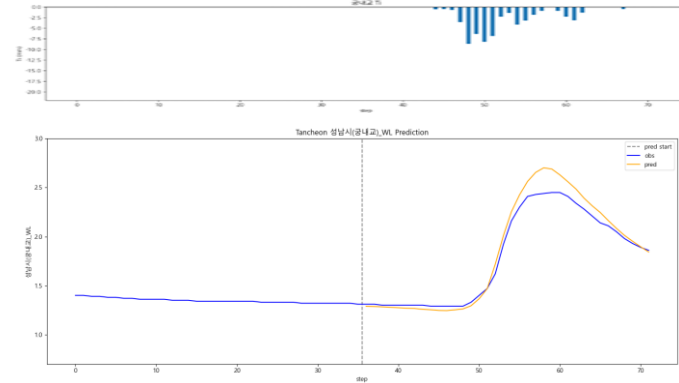
R2 : 0.6406

궁내교 - 단일 LSTM

관심 수위 이상

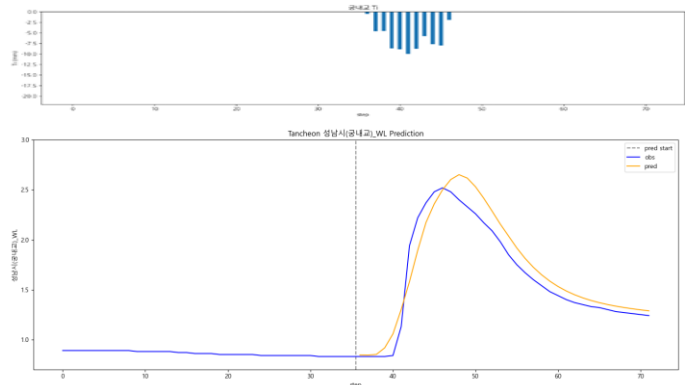
testdata_200802

R2 : 0.9508



testdata_220712

R2 : 0.0509



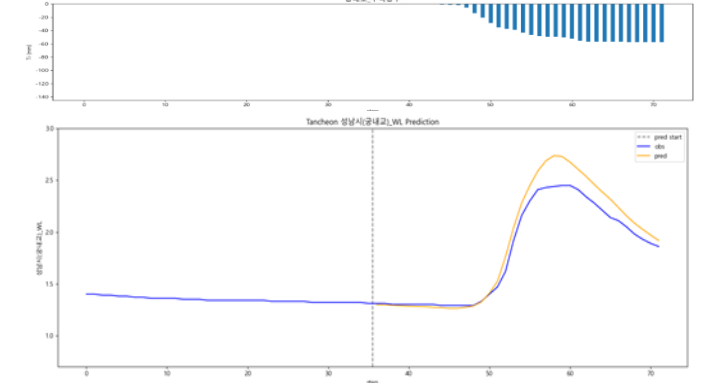
testdata_230711

R2 : 0.9096

관심 수위 이상 + 누적강우

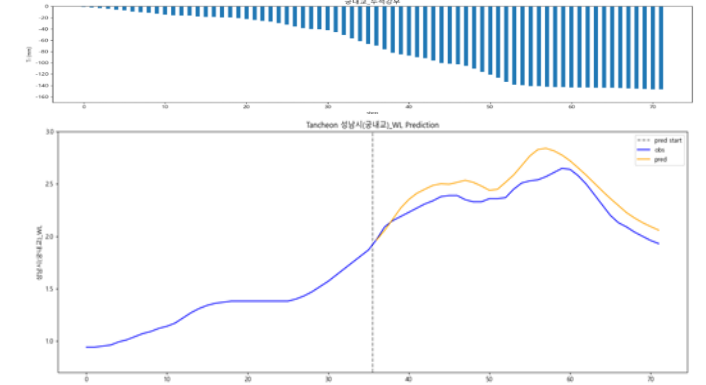
testdata_200802

R2 : 0.9176



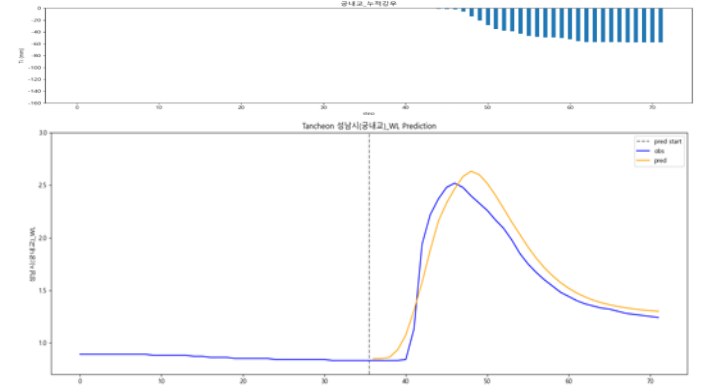
testdata_220712

R2 : 0.4645



testdata_230711

R2 : 0.9119

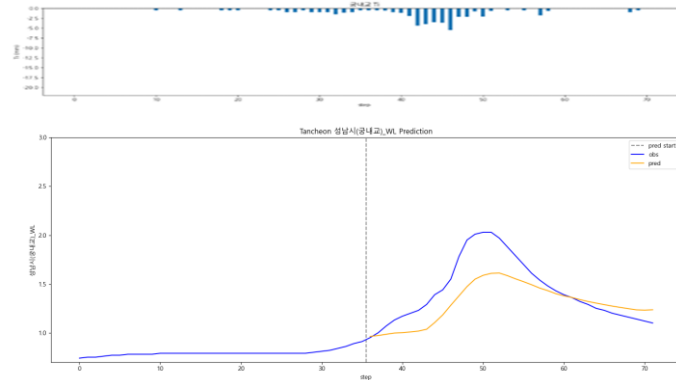
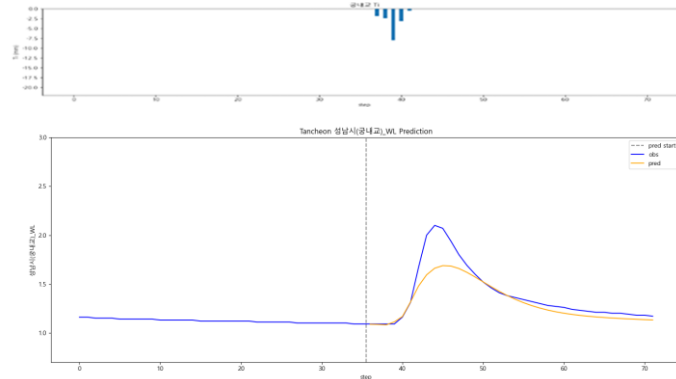


궁내교 - 단일 LSTM

관심 수위 이상

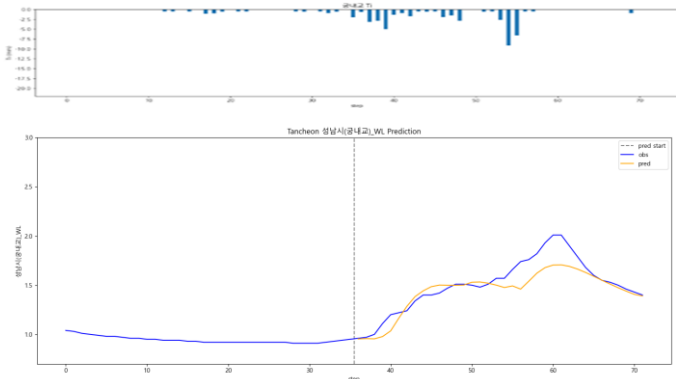
testdata_240723

R2 : 0.7689



testdata_250716

R2 : 0.5051



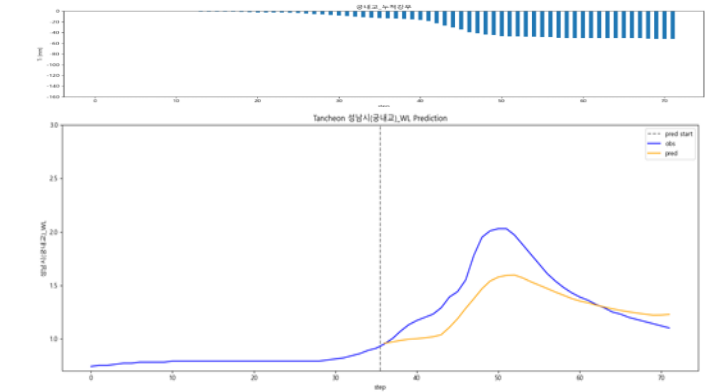
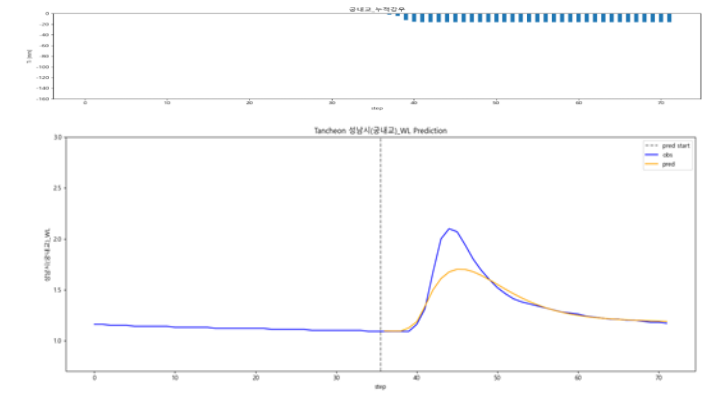
testdata_250814

R2 : 0.7659

관심 수위 이상 + 누적강우

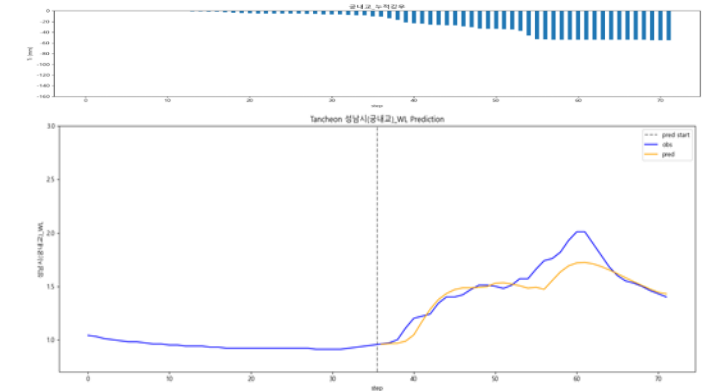
testdata_240723

R2 : 0.7989



testdata_250716

R2 : 0.4844



testdata_250814

R2 : 0.7933

2020년 이후 데이터
+ 관심수위
궁내교

2020년 이후 데이터 + 관심수위 궁내교

- 3시간 : 0.43(83.0%)
2.51 – 2.94
- 1시간 : -0.50(74.4%)
1.94 – 1.44

• Parallel LSTM

관심 수위 이상				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9513	0.0102	0.0814	0.1013
22.07.12	-0.8548	0.0735	0.2146	0.2712
23.07.11	0.8751	0.0349	0.1217	0.1868
24.07.23	0.8136	0.0150	0.0610	0.1227
25.07.16	0.8213	0.0179	0.0927	0.1338
25.08.14	0.7835	0.0152	0.0984	0.1235

관심 수위 이상 + 누적강우

* 몇시간 뒤 시점 : 오차 (유사율)

관측수위 - 예측수위

누적				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9507	0.0104	0.0669	0.1020
22.07.12	0.2329	0.0304	0.1562	0.1744
23.07.11	0.8663	0.0374	0.1237	0.1934
24.07.23	0.8524	0.0119	0.0770	0.1092
25.07.16	0.7764	0.0224	0.1197	0.1497
25.08.14	0.5660	0.0305	0.1484	0.1749

- 1시간 : -0.62(68.2%)
1.94 – 1.32
- 3시간 : -0.27(84.7%)
1.79 – 1.52

관심 수위 이상				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9610	0.0083	0.0658	0.0908
22.07.12	0.2350	0.0304	0.1573	0.1742
23.07.11	0.7765	0.0625	0.2219	0.2501
24.07.23	0.7457	0.0206	0.0937	0.1434
25.07.16	0.6535	0.0348	0.1430	0.1864
25.08.14	0.8250	0.0123	0.0870	0.1111

- 1시간 : -0.32(83.3 %)
1.94 – 1.62
- 3시간 : 0.35(81.2 %)
1.85 – 2.2
- 1시간 : -0.2(83.4 %)
1.23 – 1.03

• 단일 LSTM

누적				
궁내교				
Test data	r2	mse	mae	rmse
20.08.02	0.9407	0.0125	0.1010	0.1120
22.07.12	0.2055	0.0315	0.1429	0.1775
23.07.11	0.6441	0.0996	0.2819	0.3156
24.07.23	0.6904	0.0250	0.1010	0.1583
25.07.16	0.7551	0.0246	0.1311	0.1567
25.08.14	0.9146	0.0060	0.0616	0.0776

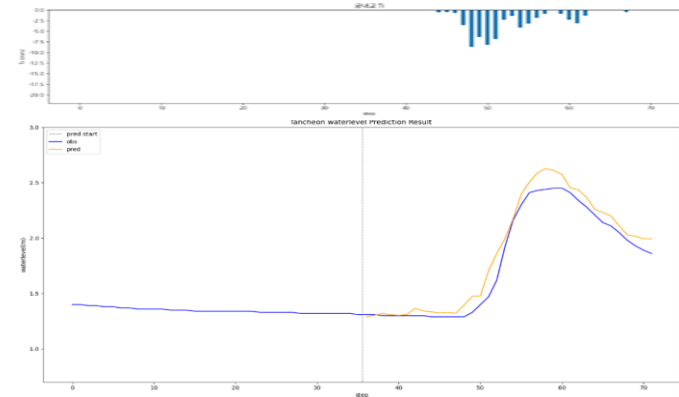
- 3시간 : 0.46(75.3 %)
1.85 – 2.31

궁내교 – Parallel LSTM

관심 수위 이상

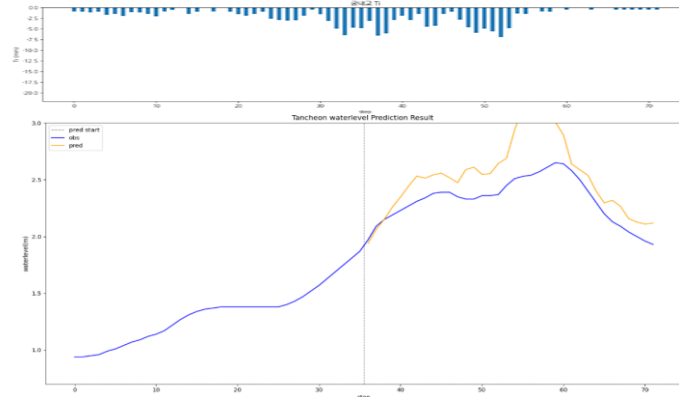
testdata_200802

R2 : 0.9513



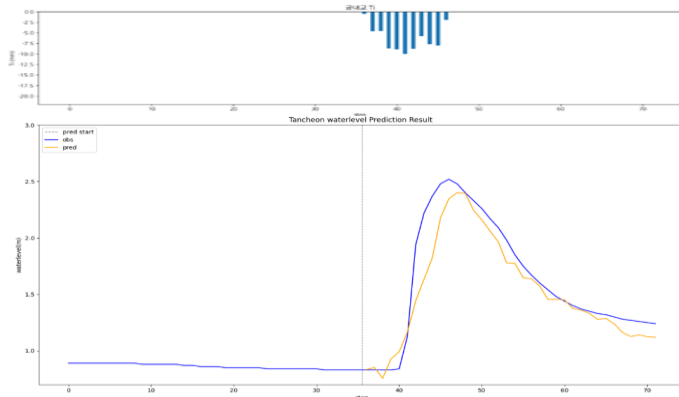
testdata_220712

R2 : -0.8548



testdata_230711

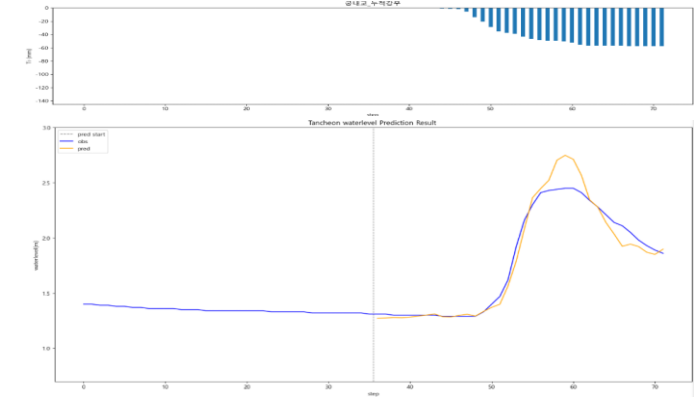
R2 : 0.8751



관심 수위 이상 + 누적강우

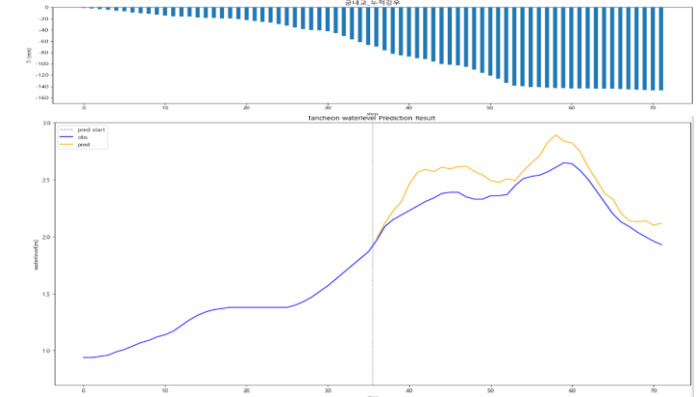
testdata_200802

R2 : 0.9507



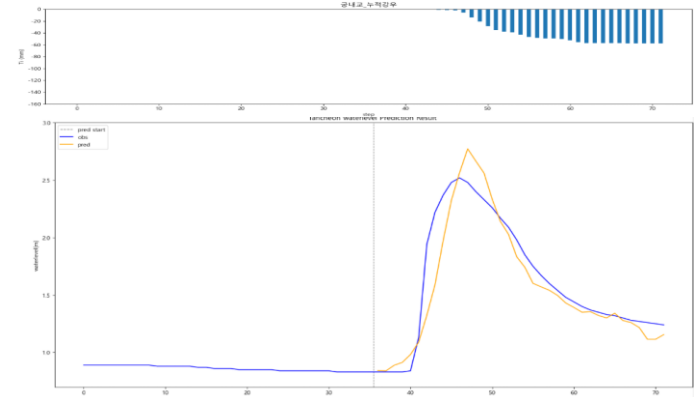
testdata_220712

R2 : 0.2329



testdata_230711

R2 : 0.8663

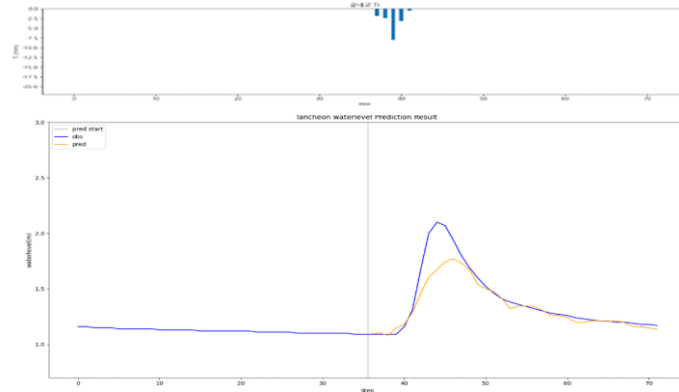


궁내교 – Parallel LSTM

관심 수위 이상

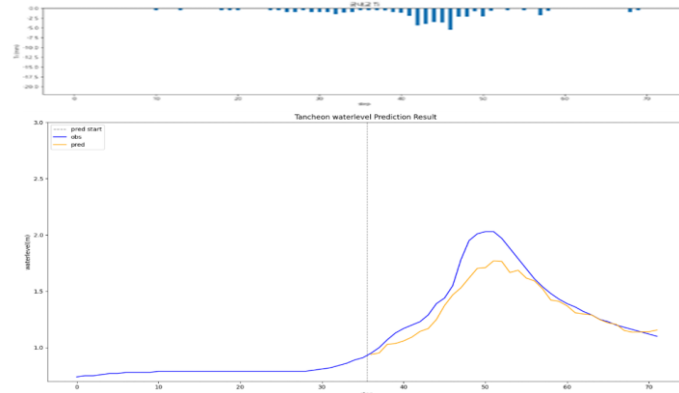
testdata_240723

R2 : 0.8136



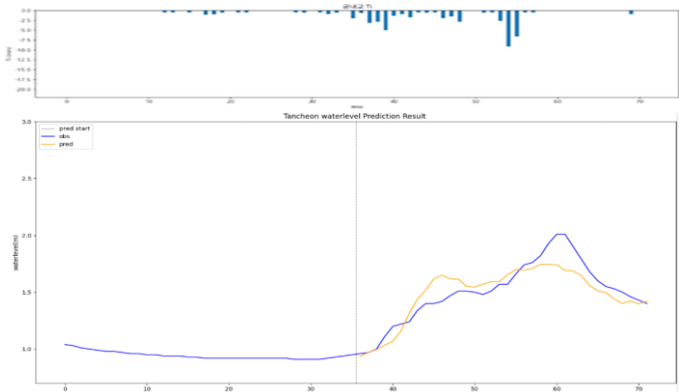
testdata_250716

R2 : 0.8213



testdata_250814

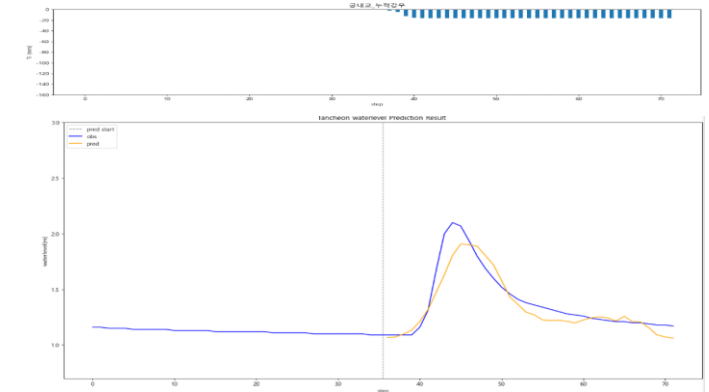
R2 : 0.7835



관심 수위 이상 + 누적강우

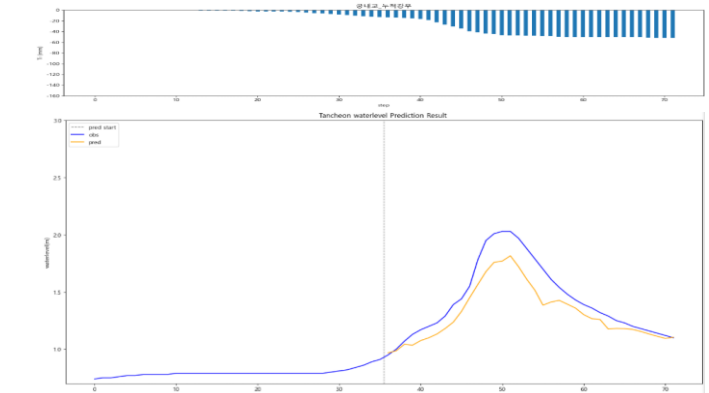
testdata_240723

R2 : 0.8524



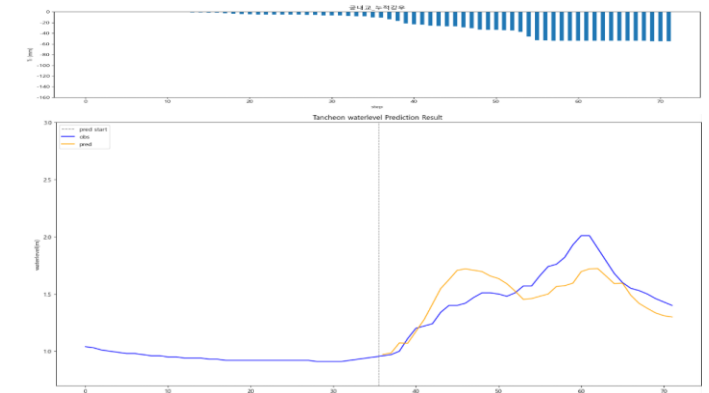
testdata_250716

R2 : 0.7764



testdata_250814

R2 : 0.5660

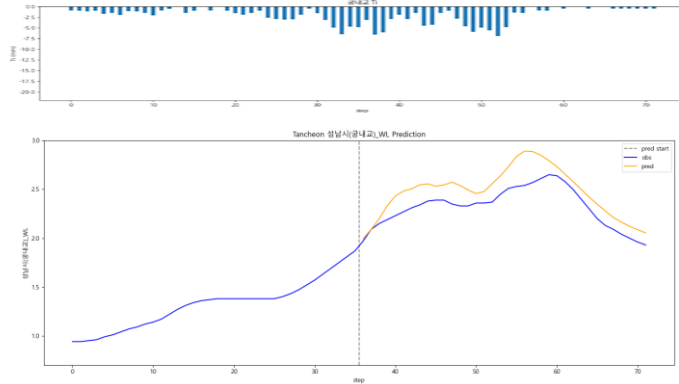
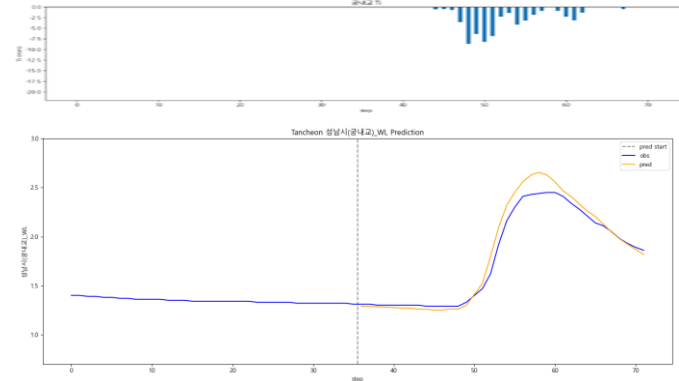


궁내교 - 단일 LSTM

관심 수위 이상

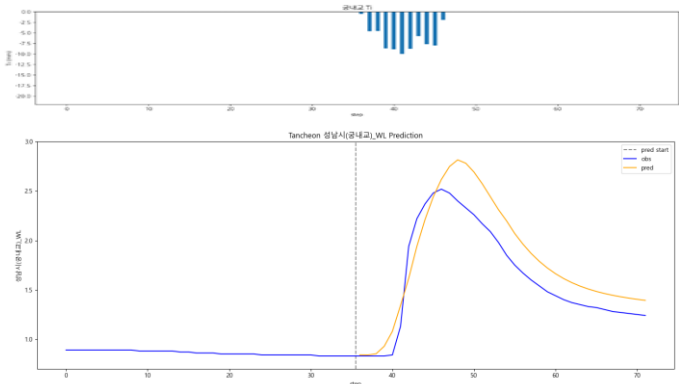
testdata_200802

R2 : 0.9610



testdata_220712

R2 : 0.2350



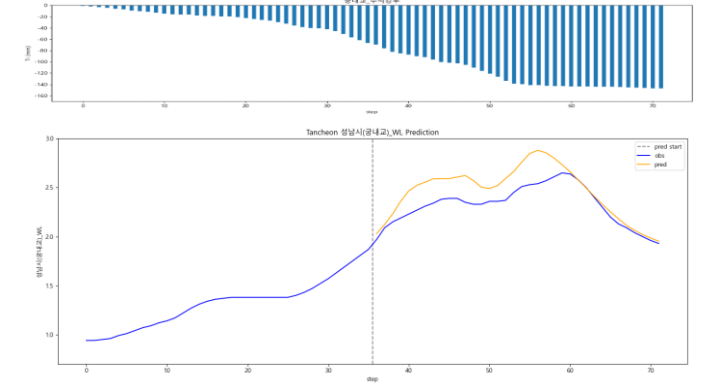
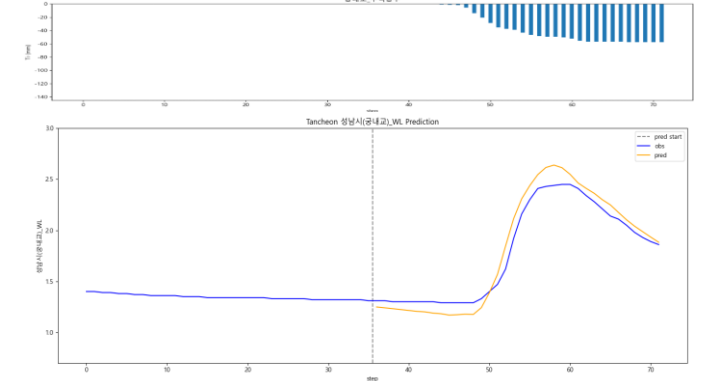
testdata_230711

R2 : 0.7765

관심 수위 이상 + 누적강우

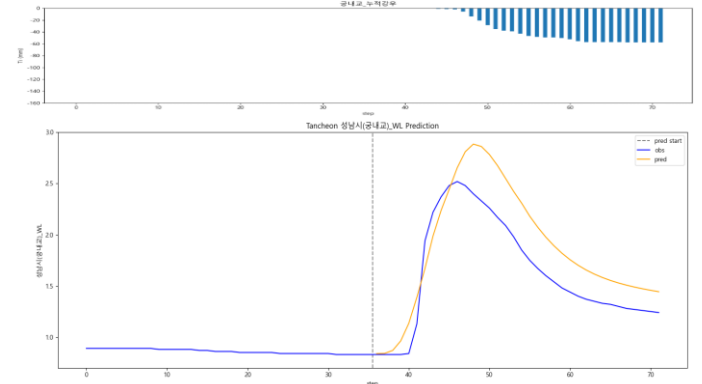
testdata_200802

R2 : 0.9407



testdata_220712

R2 : 0.2055



testdata_230711

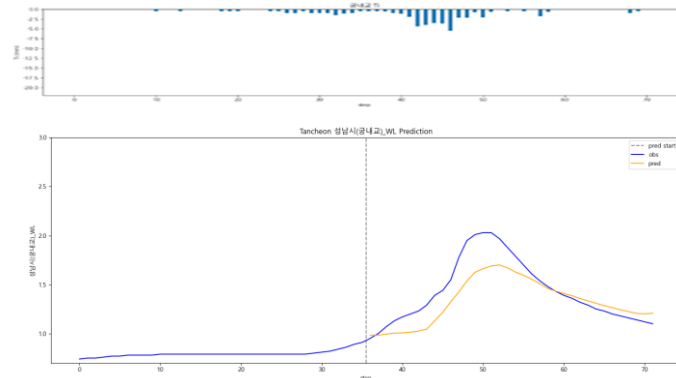
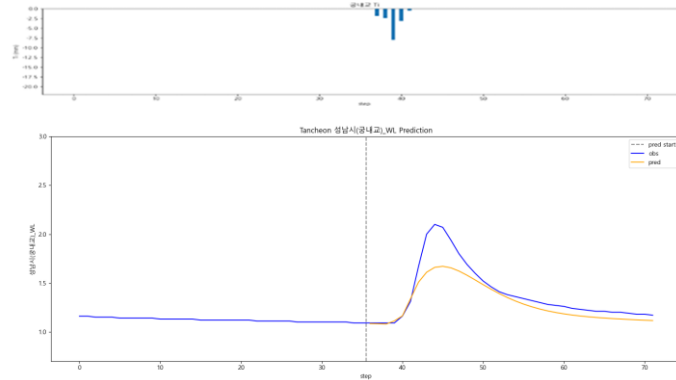
R2 : 0.6441

궁내교 - 단일 LSTM

관심 수위 이상

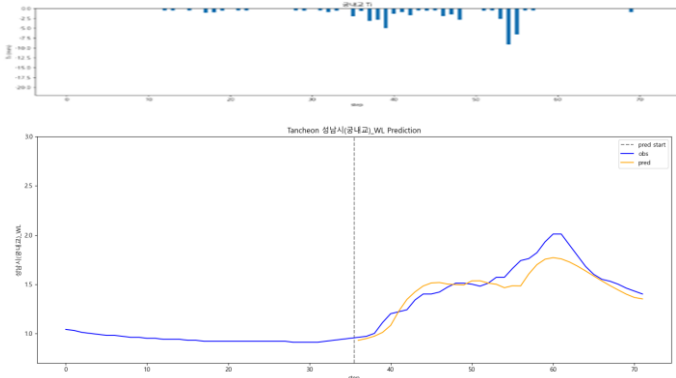
testdata_240723

R2 : 0.7457



testdata_250716

R2 : 0.6535



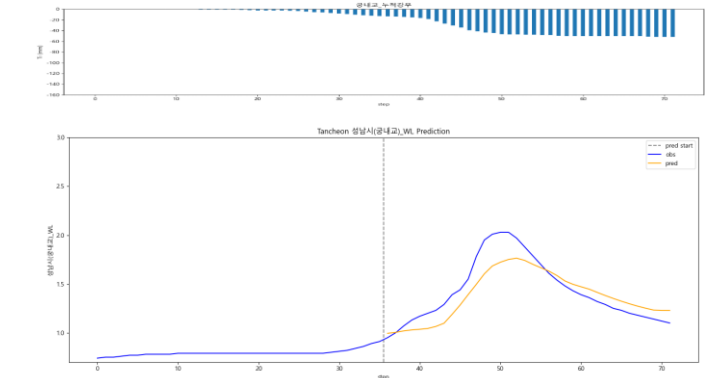
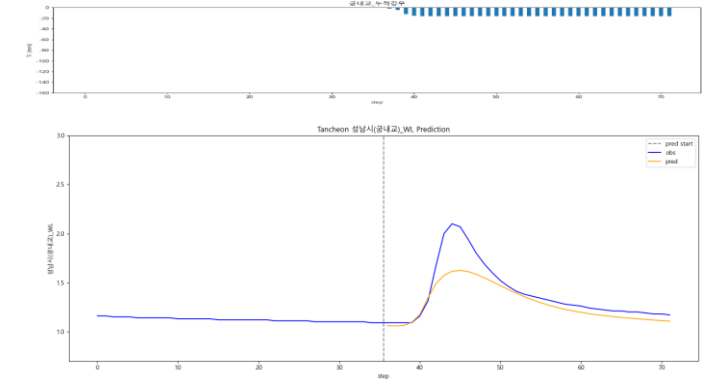
testdata_250814

R2 : 0.8250

관심 수위 이상 + 누적강우

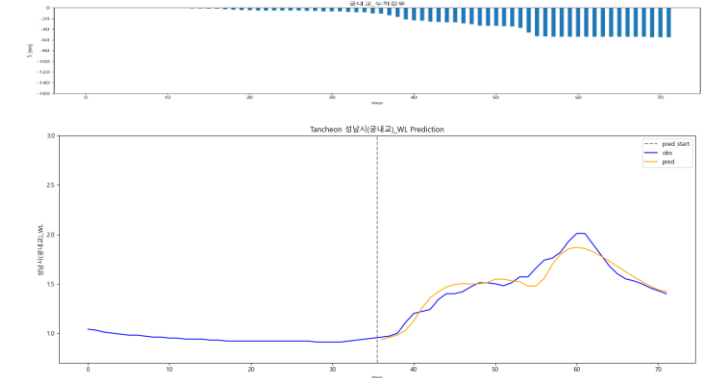
testdata_240723

R2 : 0.6904



testdata_250716

R2 : 0.7551



testdata_250814

R2 : 0.9146

관심수위
대곡교

관심수위 대곡교

- 3시간 : -0.69(80.7%)
3.6 – 2.91

Parallel LSTM

- 1시간 : -0.45(84.2%)
2.86 – 2.41

관심 수위 이상

대곡교				
Test data	r2	mse	mae	rmse
20.08.02	0.8748	0.0763	0.1754	0.2763
22.07.12	0.5503	0.0629	0.2047	0.2509
23.07.11	0.8659	0.1219	0.2258	0.3492
24.07.23	0.6749	0.0154	0.0860	0.1244
25.07.16	0.9470	0.0244	0.1098	0.1563
25.08.14	0.8781	0.0423	0.1589	0.2056

관심 수위 이상 + 누적강우

* 몇시간 뒤 시점 : 오차 (유사율)

관측수위 - 예측수위

누적 대곡교				
Test data	r2	mse	mae	rmse
20.08.02	0.7441	0.1561	0.2908	0.3951
22.07.12	0.6399	0.0504	0.1701	0.2245
23.07.11	0.8559	0.1311	0.2905	0.3621
24.07.23	0.5656	0.0206	0.1076	0.1437
25.07.16	0.9182	0.0377	0.1533	0.1942
25.08.14	0.8301	0.0589	0.1976	0.2428

- 3시간 : -0.88(75.6 %)
3.6 – 2.72

대곡교

Test data	r2	mse	mae	rmse
20.08.02	0.9090	0.0555	0.1820	0.2356
22.07.12	0.5072	0.0690	0.2390	0.2627
23.07.11	0.8436	0.1423	0.3187	0.3773
24.07.23	0.5396	0.0219	0.1230	0.1480
25.07.16	0.9218	0.0361	0.1634	0.1900
25.08.14	0.9005	0.0345	0.1388	0.1858

단일 LSTM

- 1시간 : -0.44(84.7 %)
2.86 – 2.42

누적

대곡교

Test data	r2	mse	mae	rmse
20.08.02	0.9422	0.0353	0.1479	0.1879
22.07.12	0.6228	0.0528	0.2082	0.2298
23.07.11	0.8492	0.1372	0.3123	0.3704
24.07.23	0.5213	0.0228	0.1271	0.1509
25.07.16	0.9162	0.0387	0.1629	0.1967
25.08.14	0.8923	0.0374	0.1438	0.1934

- 1시간 : -0.72(75 %)
2.86 – 2.14
- 3시간 : 0.48(83.9 %)
2.97 – 3.45

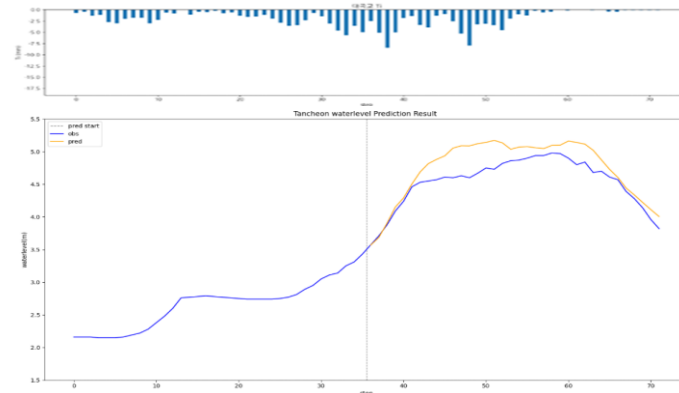
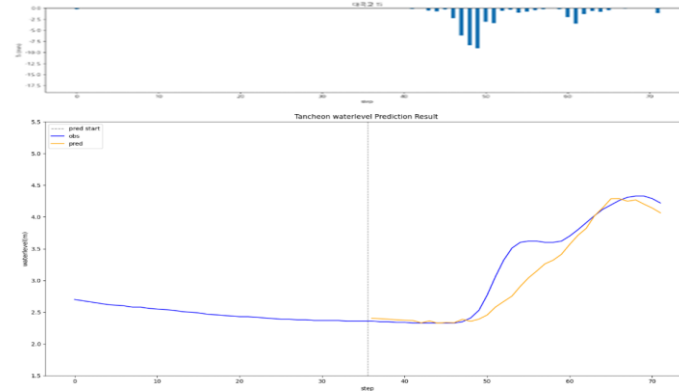
- 1시간 : 0.5(73.7 %)
1.91 – 2.41
- 3시간 : -0.77(79.5 %)
3.75 – 2.98

대곡교 - Parallel LSTM

관심 수위 이상

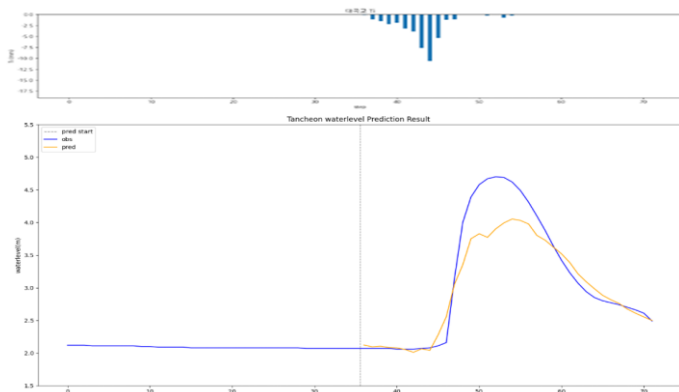
testdata_200802

R2 : 0.8748



testdata_220712

R2 : 0.5503



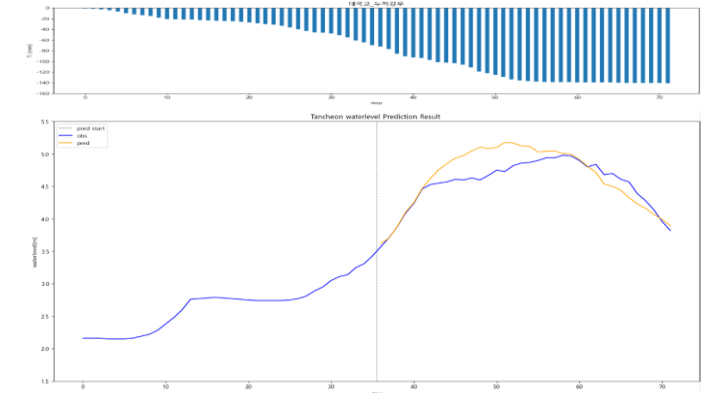
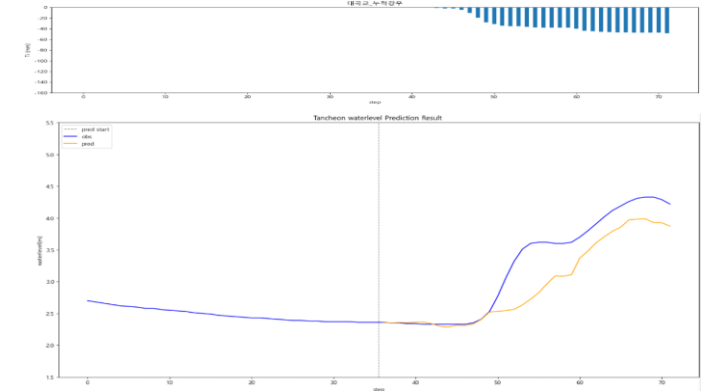
testdata_230711

R2 : 0.8659

관심 수위 이상 + 누적강우

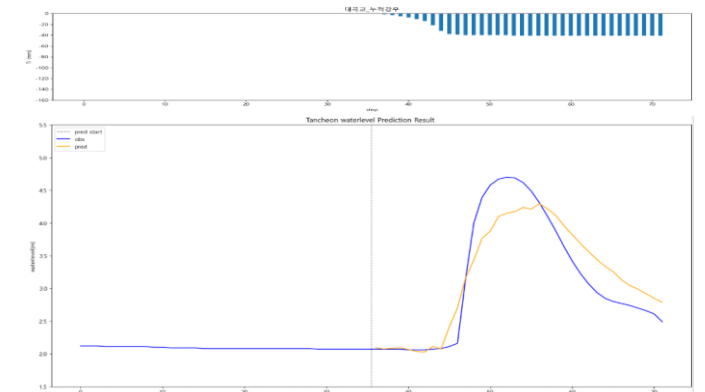
testdata_200802

R2 : 0.7441



testdata_220712

R2 : 0.6399



testdata_230711

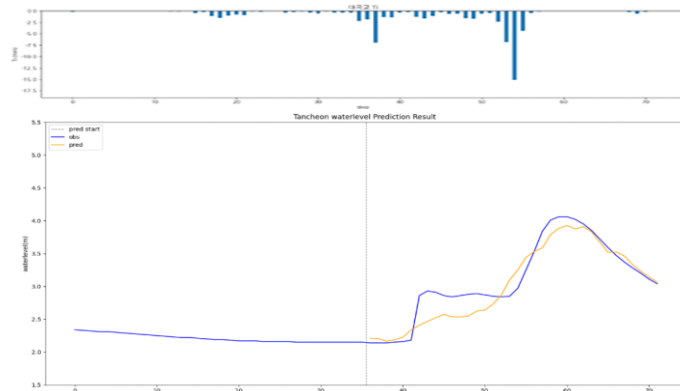
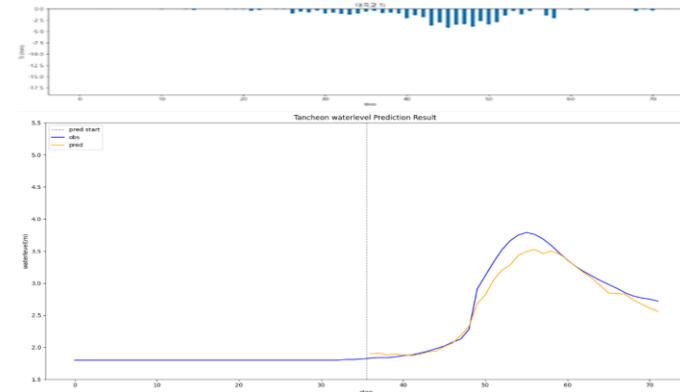
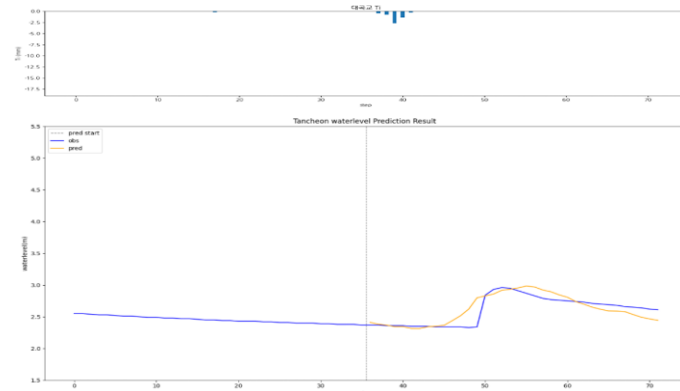
R2 : 0.8559

대곡교 - Parallel LSTM

관심 수위 이상

testdata_240723

R2 : 0.6749



testdata_250716

R2 : 0.9470

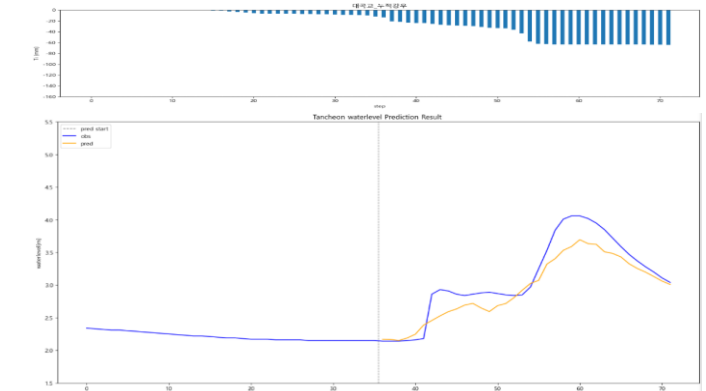
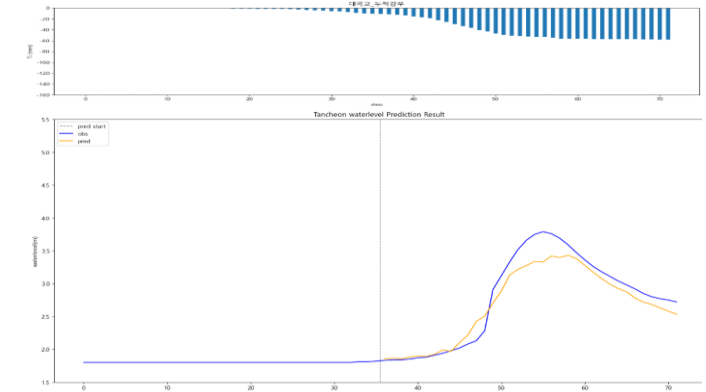
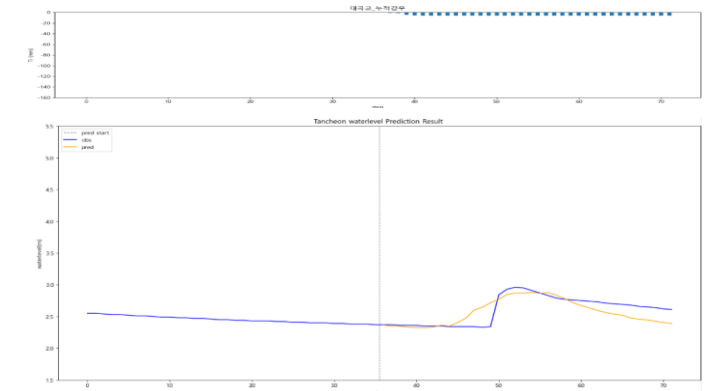
testdata_250814

R2 : 0.8781

관심 수위 이상 + 누적강우

testdata_240723

R2 : 0.5656



testdata_250716

R2 : 0.9182

testdata_250814

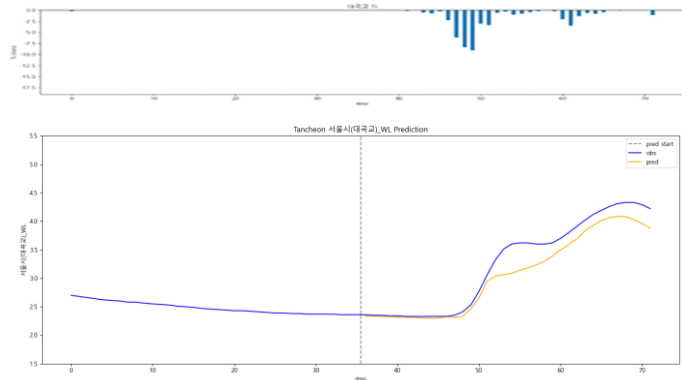
R2 : 0.8301

대곡교 - 단일 LSTM

관심 수위 이상

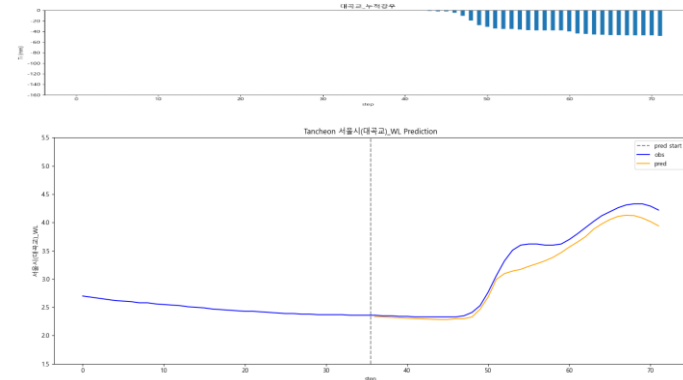
testdata_200802

R2 : 0.9090



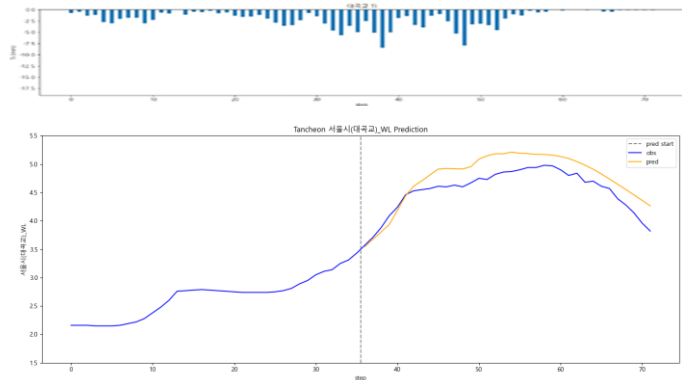
testdata_200802

R2 : 0.9422



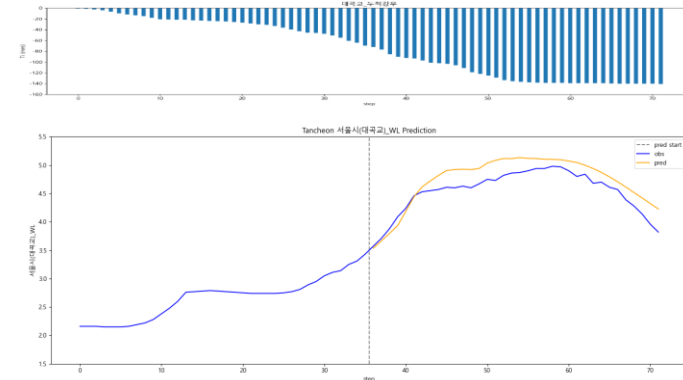
testdata_220712

R2 : 0.5072



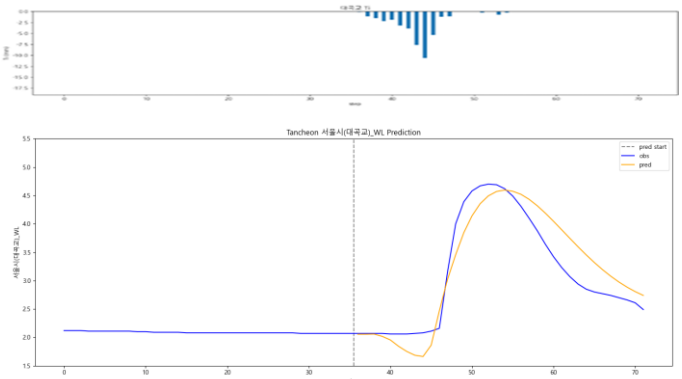
testdata_220712

R2 : 0.6228



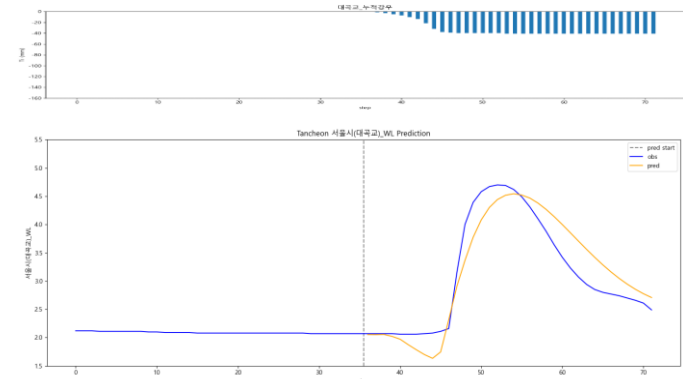
testdata_230711

R2 : 0.8436

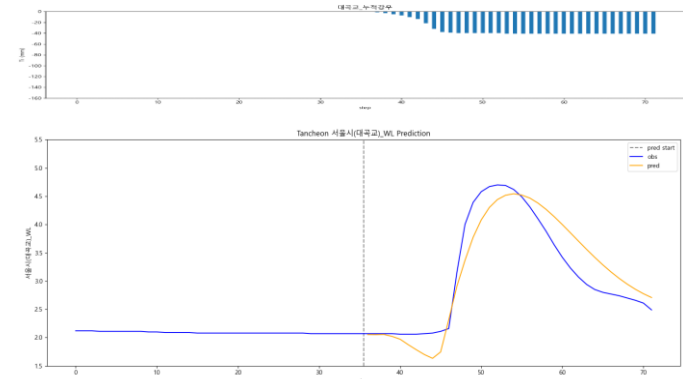
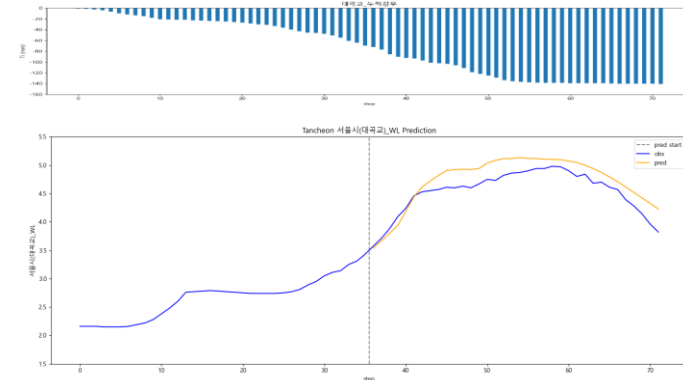
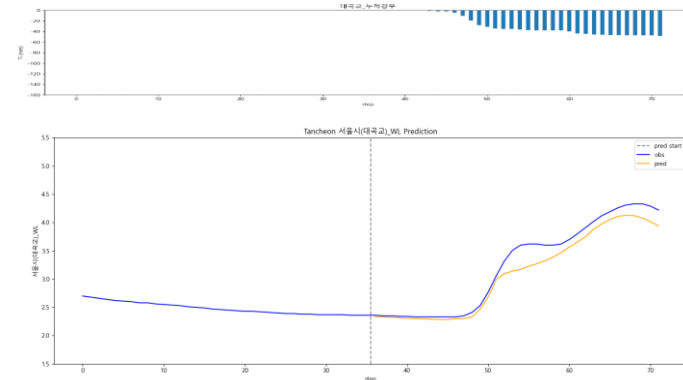


testdata_230711

R2 : 0.8492



관심 수위 이상 + 누적강우

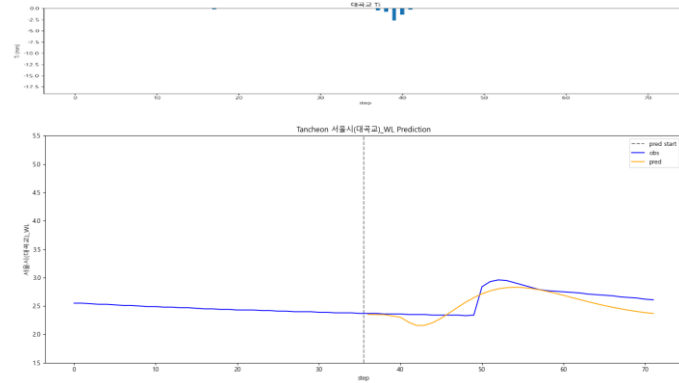


대곡교 - 단일 LSTM

관심 수위 이상

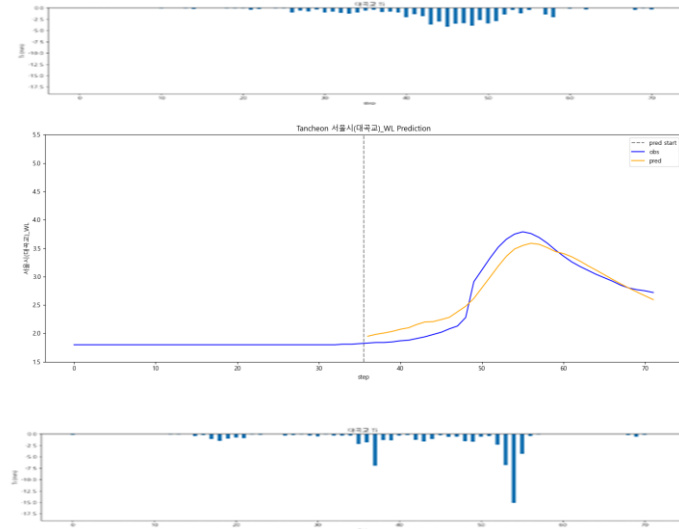
testdata_240723

R2 : 0.5396



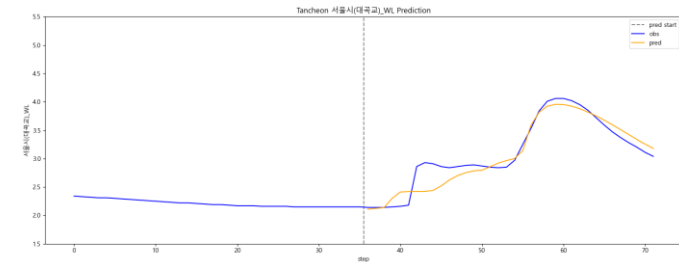
testdata_250716

R2 : 0.9218



testdata_250814

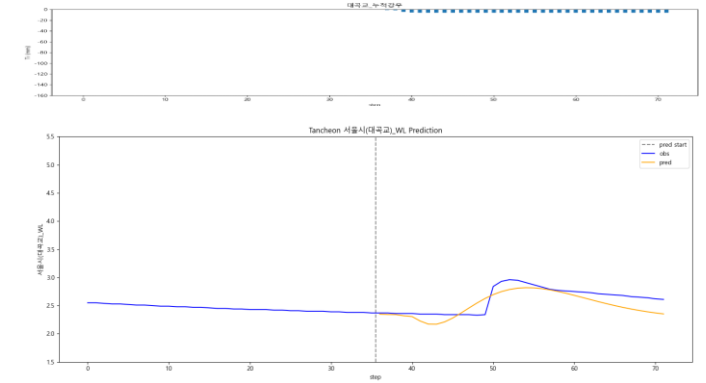
R2 : 0.9005



관심 수위 이상 + 누적강우

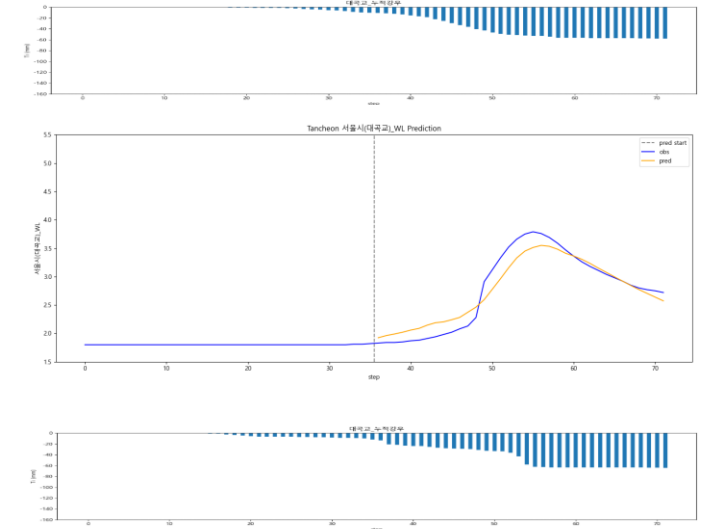
testdata_240723

R2 : 0.5213



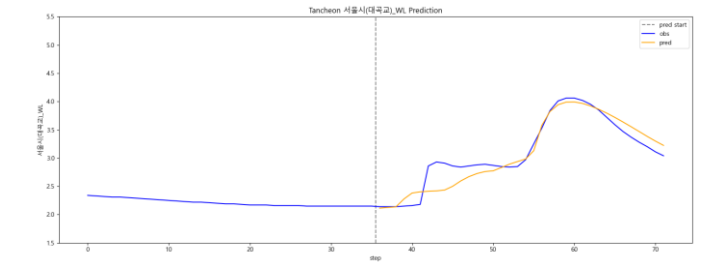
testdata_250716

R2 : 0.9162



testdata_250814

R2 : 0.8923



2020년 이후 데이터
+ 관심수위
대곡교

2020년 이후 데이터 + 관심수위 대곡교

• Parallel LSTM

• 3시간 : -0.68(81.1%)
3.6 - 2.92

• 3시간 : -0.82(82.2%)
4.62 - 3.8

• 3시간 : -0.58(84.7%)
3.75 - 3.17

관심 수위 이상

대곡교				
Test data	r2	mse	mae	rmse
20.08.02	0.8845	0.0704	0.1797	0.2653
22.07.12	-0.1561	0.1619	0.3550	0.4023
23.07.11	0.7304	0.2453	0.3701	0.4953
24.07.23	0.6077	0.0186	0.0981	0.1365
25.07.16	0.8647	0.0624	0.1862	0.2498
25.08.14	0.8840	0.0402	0.1572	0.2006

관심 수위 이상 + 누적강우

누적	대곡교			
Test data	r2	mse	mae	rmse
20.08.02	0.8523	0.0900	0.2342	0.3001
22.07.12	-0.5266	0.2137	0.4104	0.4623
23.07.11	0.7527	0.2250	0.3573	0.4743
24.07.23	0.6739	0.0155	0.0883	0.1245
25.07.16	0.9528	0.0217	0.1085	0.1475
25.08.14	0.8631	0.0474	0.1706	0.2179

• 단일 LSTM

• 1시간 : 0.31(84 %)
1.91 - 2.22

대곡교

대곡교				
Test data	r2	mse	mae	rmse
20.08.02	0.9578	0.0257	0.1122	0.1604
22.07.12	0.3842	0.0862	0.2608	0.2937
23.07.11	0.7163	0.2582	0.3972	0.5081
24.07.23	0.4315	0.0270	0.1404	0.1644
25.07.16	0.9106	0.0413	0.1770	0.2032
25.08.14	0.9172	0.0287	0.1304	0.1695

누적

대곡교

누적	대곡교			
Test data	r2	mse	mae	rmse
20.08.02	0.9719	0.0171	0.1040	0.1309
22.07.12	0.5927	0.0570	0.1987	0.2388
23.07.11	0.7324	0.2436	0.3876	0.4936
24.07.23	0.3858	0.0292	0.1499	0.1709
25.07.16	0.9196	0.0371	0.1666	0.1927
25.08.14	0.9006	0.0345	0.1452	0.1857

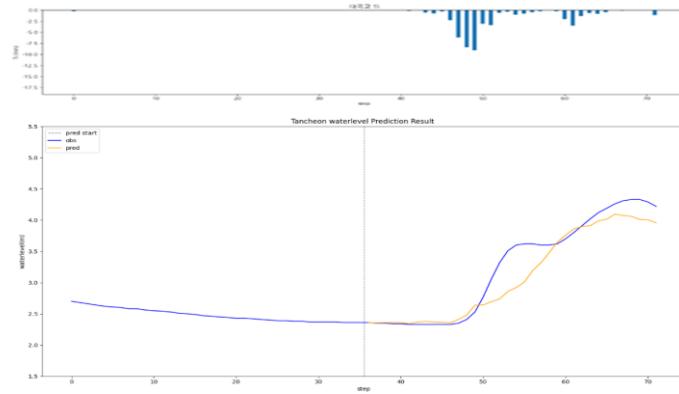
• 1시간 : 0.29(84.6 %)
1.91 - 2.2

대곡교 – Parallel LSTM

관심 수위 이상

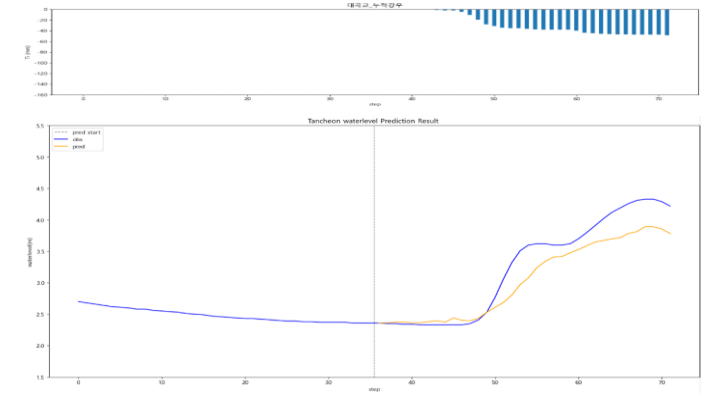
testdata_200802

R2 : 0.8845



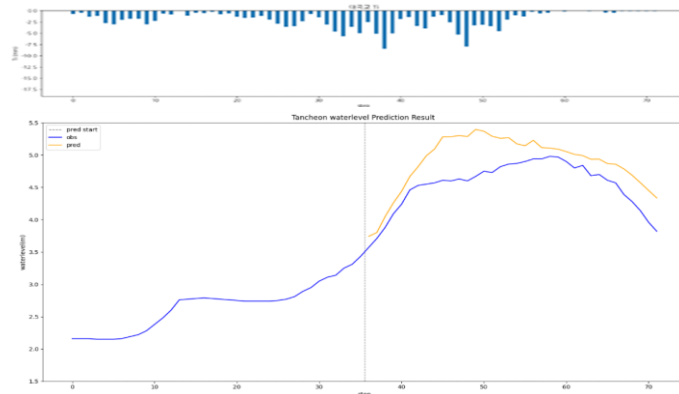
testdata_200802

R2 : 0.8523



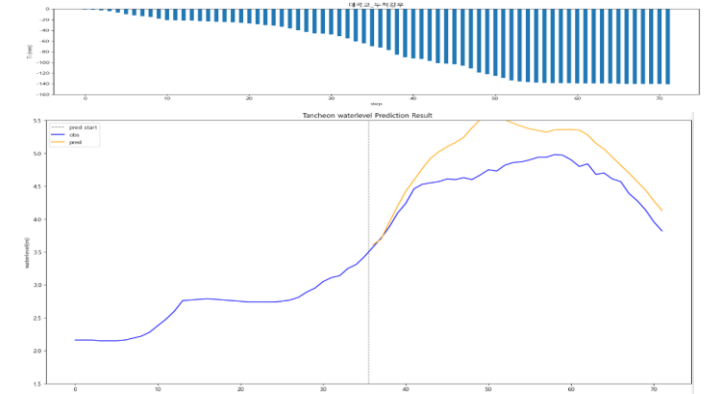
testdata_220712

R2 : -0.1561



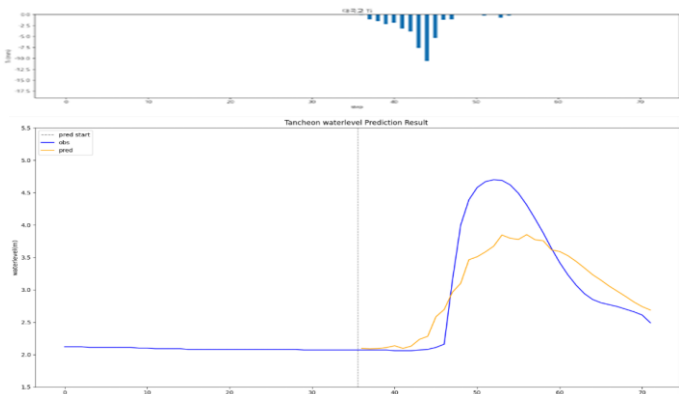
testdata_220712

R2 : -0.5266



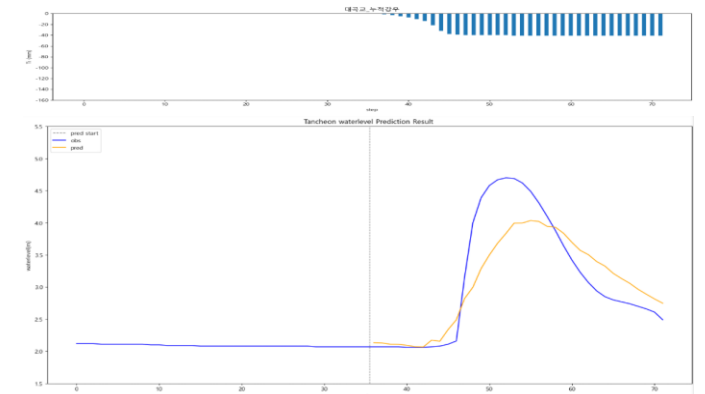
testdata_230711

R2 : 0.7304



testdata_230711

R2 : 0.7527



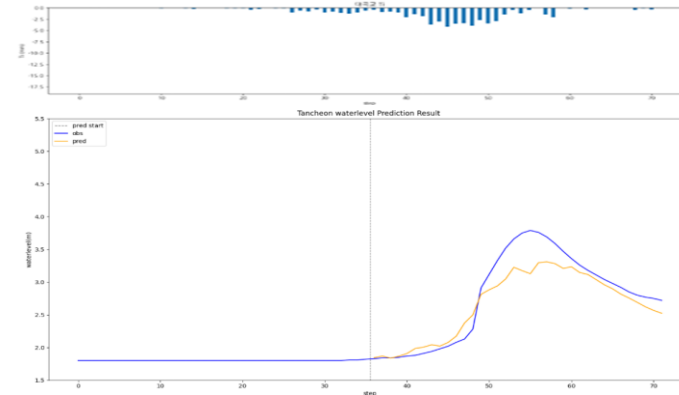
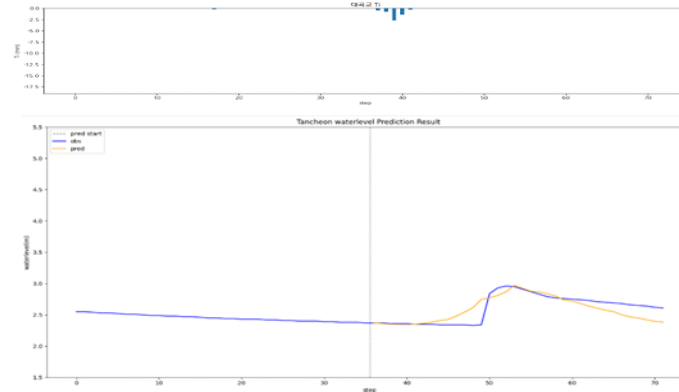
관심 수위 이상 + 누적강우

대곡교 – Parallel LSTM

관심 수위 이상

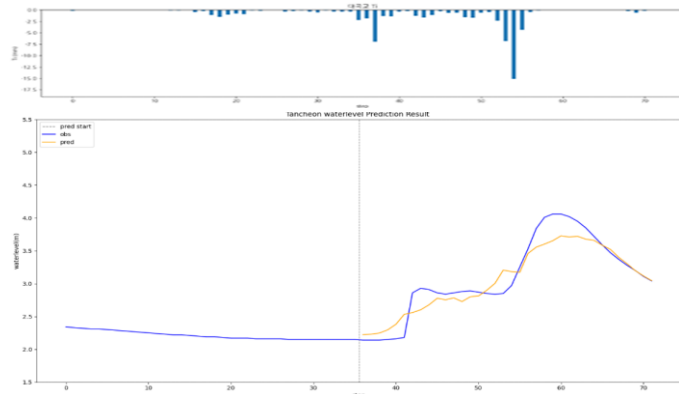
testdata_240723

R2 : 0.6077



testdata_250716

R2 : 0.8647



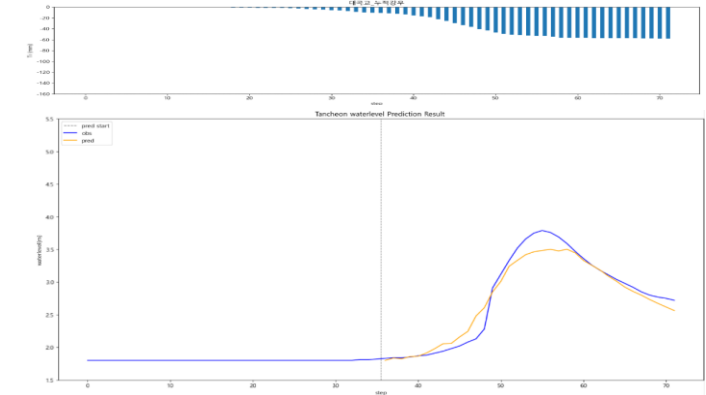
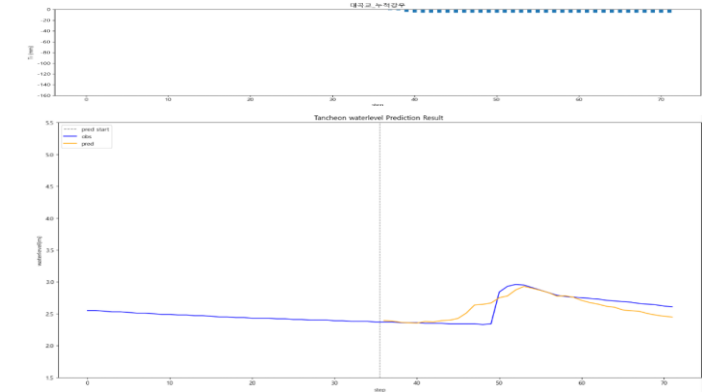
testdata_250814

R2 : 0.8840

관심 수위 이상 + 누적강우

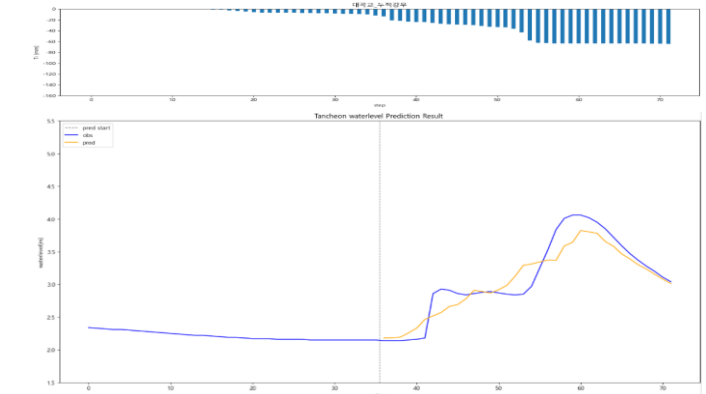
testdata_240723

R2 : 0.6739



testdata_250716

R2 : 0.9528



testdata_250814

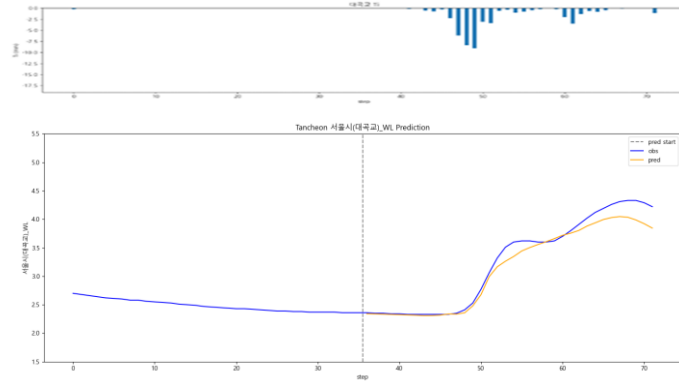
R2 : 0.8631

대곡교 - 단일 LSTM

관심 수위 이상

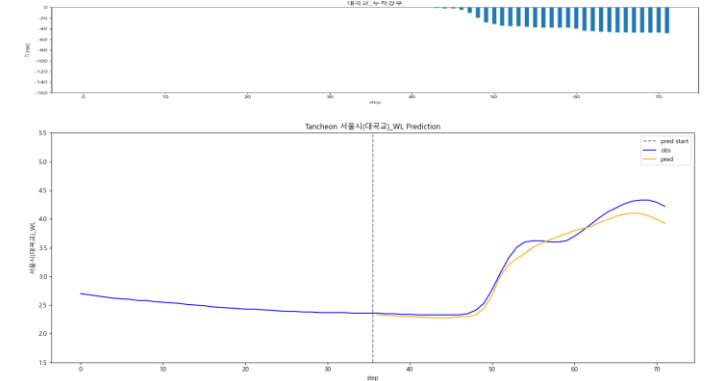
testdata_200802

R2 : 0.9578



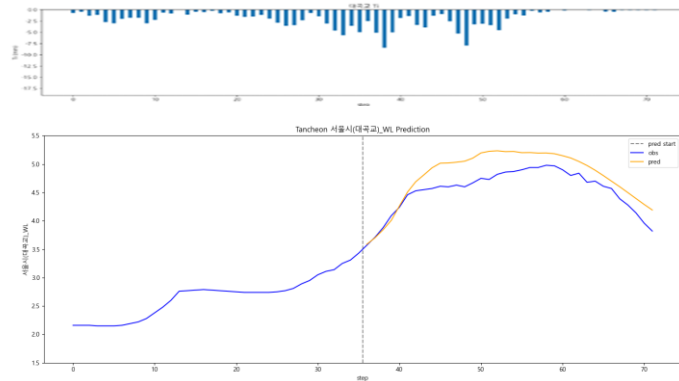
testdata_200802

R2 : 0.9719



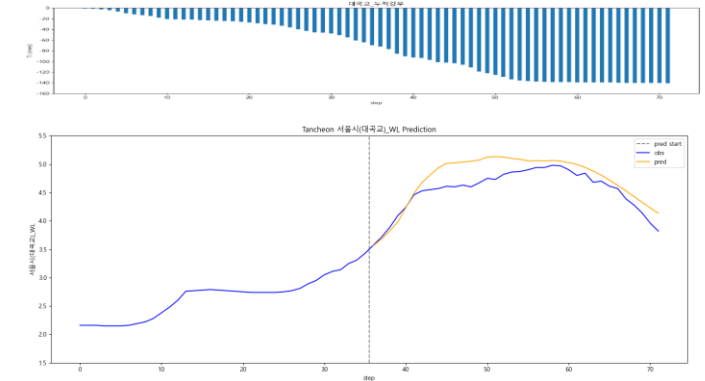
testdata_220712

R2 : 0.3842



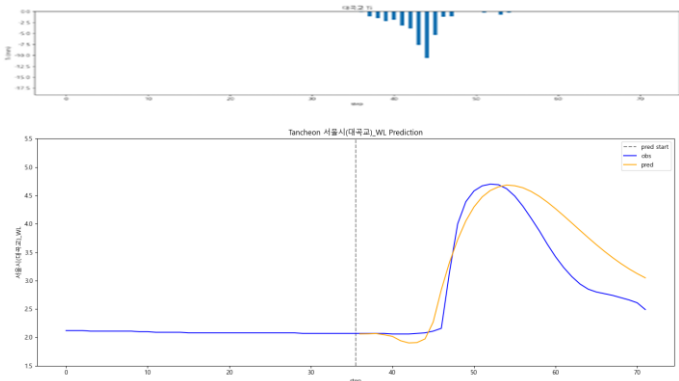
testdata_220712

R2 : 0.5927



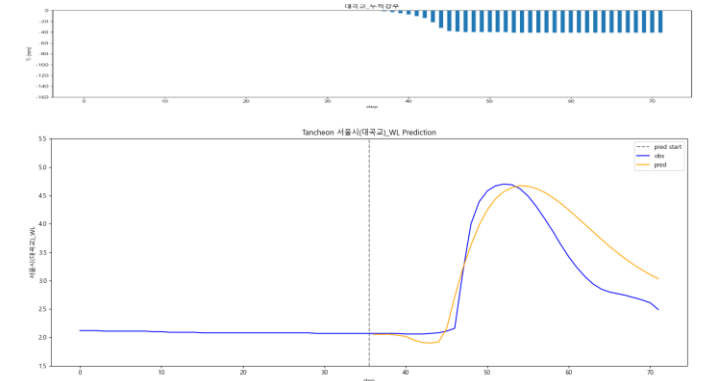
testdata_230711

R2 : 0.7163



testdata_230711

R2 : 0.7324



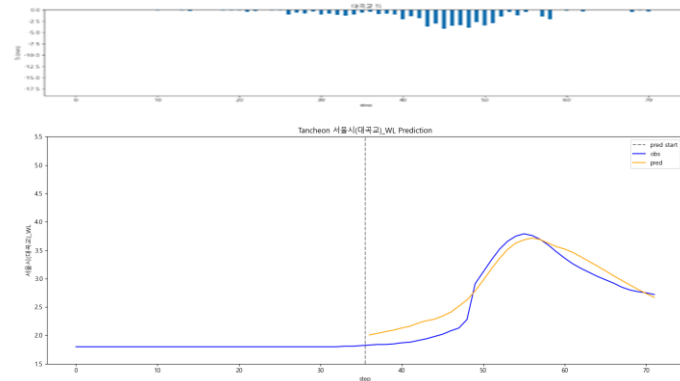
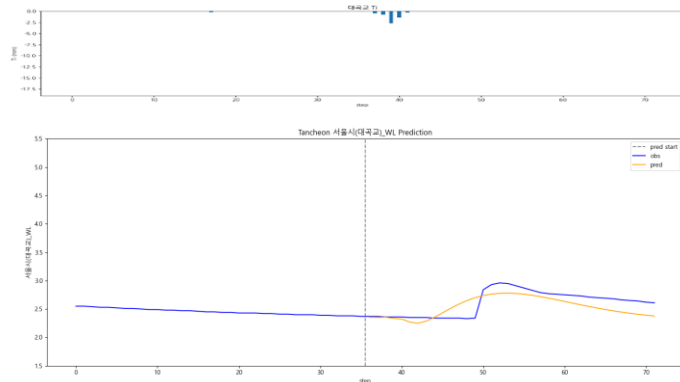
관심 수위 이상 + 누적강우

대곡교 - 단일 LSTM

관심 수위 이상

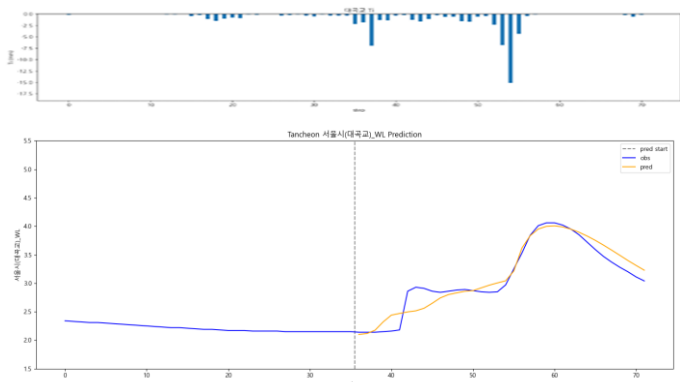
testdata_240723

R2 : 0.4315



testdata_250716

R2 : 0.9106



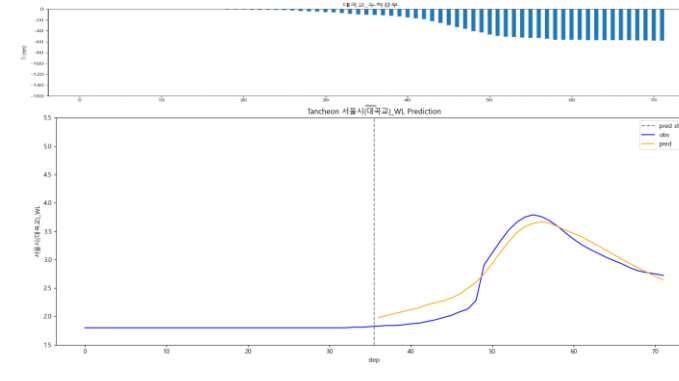
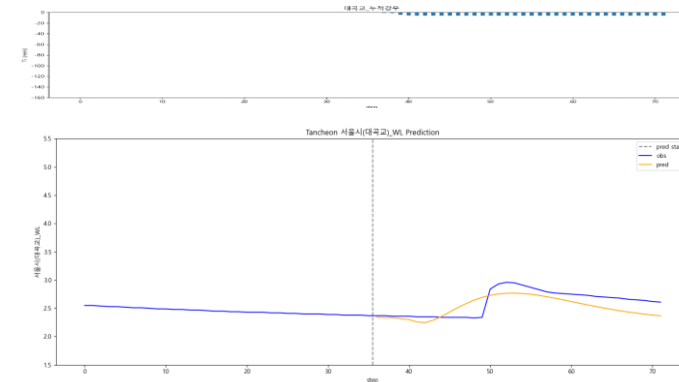
testdata_250814

R2 : 0.9172

관심 수위 이상 + 누적강우

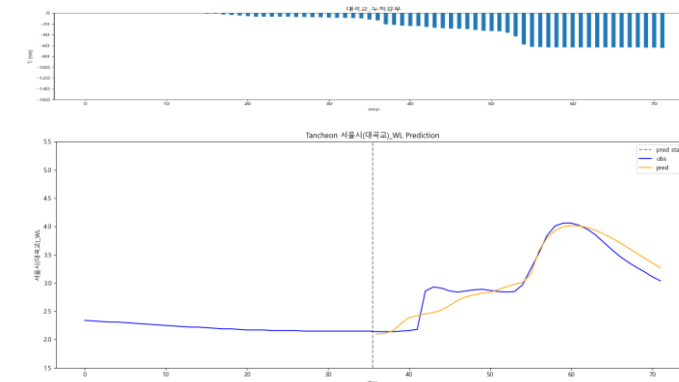
testdata_240723

R2 : 0.3858



testdata_250716

R2 : 0.9196



testdata_250814

R2 : 0.9006